



MTH205: Analysis in One Variable

Midsemester Examination, Spring 2026

Duration: 11:00AM-12:00noon.

Instructions.

- This question paper has 4 questions, each worth 5 points.
- Write your answers in the same order as the questions.
- $\mathbb{R}$ will denote the field of real numbers.
- Justify your answers clearly to obtain maximum credits. Only elegant and correct solutions will receive full credits.

Questions

- (1) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x) - f(y)| < |x - y|$ for all $x, y \in \mathbb{R}$. Prove that f must be continuous.
- (2) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. If $f(\alpha) > 0$ then prove that there is a $\delta > 0$ such that $f(x) > 0$ for all $x \in (\alpha - \delta, \alpha + \delta)$.
- (3) Let (x_n) and (y_n) be bounded real sequences. Verify that
$$\limsup(x_n + y_n) \leq \limsup(x_n) + \limsup(y_n).$$
- (4) For a set $A \subseteq \mathbb{R}$, let $\text{Cl}(A)$ denote the closure of A and $\text{int}(A)$ denote the interior of A . Prove or disprove that
$$\text{Cl}(\text{int}(A)) = \text{int}(\text{Cl}(A)).$$